NANYANG TECHNOLOGICAL UNIVERSITY

MULTIMODAL SPEECH EMOTION RECOGNITION WITH LARGE LANGUAGE MODELS

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College of Computing and Data Science

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Abstract

Speech Emotion Recognition (SER) is a critical component of human-computer interaction, with significant applications in fields such as mental health monitoring, customer service, and security surveillance. While the emergence of Large Language Models (LLMs) has demonstrated superior capabilities in semantic reasoning, applying them to SER often necessitates complex architectural adaptations. This report builds upon the SpeechCueLLM baseline, which discretizes numeric speech features into natural language descriptors to leverage the LLM's inherent reasoning abilities.

We propose an enhanced framework that integrates a structured prompting strategy, an expanded feature set, and Dynamic Feature Masking (DFM) to improve both performance and robustness. Unlike the baseline, we utilize the extended Geneva Minimalistic Acoustic Parameter Set (eGeMAPS) extracted via the openSMILE toolkit. To handle the high dimensionality of 88 acoustic parameters, we employ a hybrid feature selection strategy combining Lasso Regression and ANOVA, mapping the selected features into discrete, quantile-based natural language bins.

The framework is rigorously evaluated on two distinct domains: the acted, dyadic interactions of IEMOCAP and the naturalistic, "in-the-wild" speech of MSP-Podcast. On IEMOCAP, the model achieves an Unweighted Accuracy (UA) of 70.9%, representing a 0.3% improvement over the SpeechCueLLM baseline. On the more challenging MSP-Podcast dataset, the framework achieves a substantial 4.0% increase, reaching a Macro-F1 score of 22.5% and demonstrating competitive performance in recognizing fine-grained emotional states.

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Chapter 1

Introduction

1.1 Background

Emotion Recognition (ER) is a pivotal component of Human-Computer Interaction (HCI), enabling systems to perceive and respond to user affective states. This technology has broad downstream applications, particularly in mental health monitoring [1], automated customer service [2], and security surveillance [3]. For instance, Gaikwad and Venkatesan [4] proposed a methodology to assess depression by analyzing emotional and paralinguistic cues embedded in spoken language. Similarly, in the customer service domain, ER systems are employed to evaluate satisfaction by dynamically recognizing user sentiment [2], while in healthcare, such systems support the non-invasive detection of neurodegenerative conditions like Alzheimer's disease [5].

Human emotion is inherently multimodal, conveyed through diverse channels including speech, textual content, and facial expressions [6]. While physiological signals such as Electroencephalogram (EEG) brain waves serve as an alternative source for affective analysis [7], the primary non-invasive modalities remain acoustic and linguistic. In the acoustic domain, prosodic and voice-quality features—such as pitch, intensity, and formants—are widely utilized to capture paralinguistic information [8]. Complementing this, linguistic features derived from textual transcripts provide necessary semantic context regarding the speaker's intent and the situational causality of the emotion.

Speech Emotion Recognition (SER) is formally defined as the computational task of identifying and categorizing human emotional states directly from these speech signals. Recently, driven by the rapid proliferation and emergent reasoning capabilities of Large Language Models (LLMs), there has been a paradigm shift towards adapting these generative models for affective computing tasks. Consequently, this research focuses on **Multimodal SER**, integrating both acoustic and linguistic modalities within an LLM-based framework to achieve robust and context-aware inference.

1.2 Challenges and Limitations

ER remains a challenging endeavor due to the inherent ambiguity and complexity of human affect. Ahmed et al. [9] observe that unimodal models often struggle to capture the comprehensive emotional state of a user, thereby necessitating the adoption of multimodal approaches. However, the non-linear interaction between these modalities introduces significant computational complexity [9]. Furthermore, while Deep Neural Networks (DNNs) excel at modeling these complex multimodal interactions [10], their ”black-box” nature results in a lack of explainability. This opacity severely hinders their applicability in sensitive domains such as healthcare, where interpretable decision-making is critical [11].

1.2.1 Limitations of Text Emotion Recognition (TER)

While textual content provides direct semantic information, relying solely on linguistic transcripts is often insufficient for accurate emotion detection. TER models frequently struggle with tonal ambiguity, failing to detect sarcasm, irony, or subtext where the spoken intonation contradicts the literal semantic meaning [12].

1.2.2 Limitations of Speech Emotion Recognition (SER)

Current approaches in SER face two primary limitations regarding context and emotional granularity:

Lack of Contextual Awareness: Many State-of-the-Art (SOTA) models, such as

emotion2vec [13], perform evaluation at the level of a single utterance. This approach overlooks the temporal dynamics of conversation, failing to account for how emotions transition or intensify over the course of a dialogue [14].

Limited Emotion Taxonomy: A significant limitation of publicly available models and datasets is their exclusive focus on coarse-grained, primary emotions (e.g., Happy, Sad, Angry) [13, 15]. This restricts their effectiveness in naturalistic, real-world scenarios—such as podcasts or therapeutic sessions—where subtle secondary states like 'Concerned' or 'Disappointed' are often more prevalent than exaggerated primary emotions [16].

1.3 Objective

The primary aim of this project is to develop a robust, fully automated Multimodal SER framework capable of processing raw audio inputs for real-world applications. Unlike traditional systems that rely on manual transcription or pre-processed feature sets, the proposed model functions as an automated pipeline—converting raw audio into structured prompts for Large Language Model (LLM) inference without human intervention. The framework seeks to bridge the gap between studio-recorded benchmarks and naturalistic scenarios (e.g., mental health monitoring) by recognizing a broad taxonomy of emotions, ranging from primary categories to subtle secondary states.

The specific research objectives are as follows:

- **To Develop a Context-Aware Dialogue System:** To enable the analysis of long-form audio conversations by integrating historical context into the decision-making process. This ensures the model can resolve ambiguity in individual utterances by conditioning predictions on the preceding dialogue flow.
- **To Design an Architecture-Agnostic Framework:** To formulate a flexible prompting strategy that allows the SER system to adapt to various datasets and emotion taxonomies (primary or secondary) without requiring structural modifications to the underlying LLM architecture.

- **To Achieve Cross-Corpus Generalization:** To validate system robustness by training on studio-acted data (e.g., IEMOCAP [15]) and evaluating zero-shot performance on naturalistic, out-of-distribution datasets (e.g., MSP-Podcast [16]), thereby ensuring that learned features are transferrable rather than dataset-specific.
- **To Benchmark against State-of-the-Art (SOTA):** To quantitatively evaluate the proposed framework against existing off-the-shelf baselines—including both unimodal (audio-only) and multimodal approaches—to demonstrate the efficacy of LLM-driven reasoning in emotion recognition.

1.4 Project Scope

This research frames SER as a **closed-set classification task** leveraging the generative capabilities of LLMs. The core methodology utilizes an LLM to infer and generate specific emotion labels (e.g., 'Anger', 'Sadness') based on structured text prompts derived from extracted acoustic and linguistic features.

The study is subject to the following boundaries:

- **Modality Focus & Prioritization:** While the system incorporates textual transcripts, the research prioritizes **Multimodal SER**. The acoustic modality (prosody, pitch, voice quality) is treated as a critical decision-making component to resolve semantic ambiguity.
- **Language Constraints:** The scope is strictly limited to the **English language**. All pre-trained models and datasets utilized are English-based; cross-lingual emotion transfer is not covered.
- **Data Domain:** The experimental evaluation focuses on two distinct acoustic domains:
    1. **Studio/Acted Speech:** Represented by the **IEMOCAP** dataset [15], characterized by high-fidelity audio and dyadic interactions.
    2. **Naturalistic/Spontaneous Speech:** Represented by the **MSP-Podcast**

dataset [16], characterized by real-world background noise and natural conversational flows.

Out-of-Scope

The following areas are explicitly excluded from the current study:

- **Open-Set Emotion Generation:** While LLMs are capable of generating open-ended natural language descriptions of emotions, this project restricts the model to a **predefined taxonomy** of emotion classes. The model is not evaluated on its ability to generate novel or zero-shot emotion labels outside of this target set.
- **Visual Modality:** Although human emotion is inherently multimodal, this research excludes visual cues such as facial expressions and gestures. The focus is strictly placed on **auditory (paralinguistic)** and **linguistic (textual)** modalities.
- **Continuous Dimensional Regression:** A common alternative in SER is predicting continuous values for Valence, Arousal, and Dominance (VAD) [17]. This project does not frame the task as a regression problem. Instead, VAD values are utilized as **input features** within the prompt to aid the LLM's reasoning process, rather than serving as the target output.

1.5 Contributions

]This research advances the field of Multimodal SER by proposing a robust, LLM-driven framework capable of operating in naturalistic acoustic environments. The primary contributions of this work are summarized as follows:

- **Development of an End-to-End Multimodal Framework:** We propose a fully automated pipeline that eliminates the reliance on manual annotations (transcriptions and speaker metadata) required by prior baselines. By integrating automatic speech recognition and gender classification modules, the system enables practical, real-time inference from raw audio inputs.
- **Structured Prompting with Extended Acoustic Representation:** We introduce

a novel prompting strategy that replaces vague natural language descriptors with a structured feature table. Unlike the baseline's limited prosodic set, our framework incorporates the eGeMAPS feature set. We employ a hybrid feature selection mechanism (combining Lasso Regression and ANOVA) to identify up to 30 most discriminative parameters, enhancing the LLM's ability to resolve semantic ambiguity.

- **Dynamic Feature Masking (DFM) Strategy:** To mitigate overfitting and encourage robust cross-modal reasoning, we implement a Dynamic Feature Masking technique. This regularization strategy incrementally masks acoustic tokens during training, forcing the model to learn redundant feature correlations and preventing over-reliance on any single modality.

1.6 Report Structure

This report consists of five chapters:

- **Chapter 1: Introduction** - Provides an overview of the research landscape, identifying the limitations of current SER systems and establishing the motivations, objectives, and scope of the proposed multimodal LLM-based framework.
- **Chapter 2: Literature Review** - Surveys the evolution of multimodal SER, ranging from traditional acoustic feature analysis to SOTA Deep Learning approaches, with a specific focus on recent advancements in LLMs and prompt engineering.
- **Chapter 3: Methodology** - Deep dive to the baseline methodology, highlight its limitations, and details the theoretical design of the proposed framework, explaining the feature extraction pipeline, the prompt construction strategy, and the architectural integration of audio and text for Llama-3 inference.
- **Chapter 4: Experimental Setup** - Describes the implementation details, including the preprocessing of IEMOCAP and MSP-Podcast datasets, the specific hyperparameter configurations for fine-tuning, and the evaluation metrics used.

- **Chapter 5: Result Analysis** - Presents the quantitative and qualitative findings of the study, offering a critical analysis of the model's performance across different datasets, emotion classes, and modality combinations.
- **Chapter 6: Conclusion & Future Discussion** - Summarizes the key contributions and findings of the research, discusses the limitations encountered, and proposes potential directions for future improvements in multimodal SER.

Chapter 2

Literature Review

2.1 Speech Emotion Recognition (SER)

SER is inherently an interdisciplinary domain, converging linguistics, psychology, and computer science. Early approaches addressed the task using probabilistic Gaussian Mixture Models (GMMs). Subsequently, Hidden Markov Models (HMMs) were employed to better model the temporal dynamics inherent in speech [18, 19]. These traditional paradigms relied heavily on extensive feature engineering and selection, focusing primarily on low-level descriptors such as pitch, energy, and voice quality [20]. Consequently, Support Vector Machines (SVMs) became widely adopted during this era due to their effectiveness in handling high-dimensional, hand-crafted acoustic features [21].

The field later shifted toward Deep Learning (DL) algorithms, which dominate current research. Unlike traditional methods, DL techniques require minimal feature engineering, instead relying on large-scale datasets to learn representations automatically. Notable architectures include Recurrent Neural Networks (RNNs) [22] and Convolutional Neural Networks (CNNs) [23]. The introduction of attention mechanisms further advanced the field; for instance, Xie et al. [24] utilized attention-based LSTMs to weight the importance of specific speech frames dynamically. More recently, Self-Supervised Learning (SSL) has emerged as the state-of-the-art paradigm. Models such as Wav2Vec

2.0 (W2V2) [25] are pre-trained on vast unlabeled corpora and subsequently fine-tuned on task-specific datasets like IEMOCAP [15], yielding robust performance even with limited labeled data [13].

Despite these advancements, a prevailing limitation remains: the majority of existing SER models operate on a single-utterance basis, ignoring the broader conversational context.

2.2 Text Emotion Recognition (TER)

TER classifies emotions based solely on textual data, utilizing Natural Language Processing (NLP) models to interpret semantic content. Bidirectional Encoder Representations from Transformers (BERT) has become a cornerstone architecture in this domain due to its superior capacity for contextual understanding [26]. For instance, Kumar and Raman [27] utilized BERT to extract word embeddings, which were subsequently processed through a dual-channel architecture: a CNN network to extract local features and a BiLSTM network to capture sequential dependencies.

Unlike traditional SER, which often treats utterances in isolation, detecting emotions in conversational contexts is a central focus in TER. To address this, Majumder et al. [14] designed an RNN-based system capable of distinguishing between speakers (speaker modeling) to better track emotional shifts across a dialogue.

2.3 Multimodal Emotion Models

Recent research increasingly combines audio and text modalities to enhance emotion recognition performance. To leverage both modalities simultaneously, their respective feature sets must be effectively integrated. Common fusion strategies include feature-level fusion (early fusion) [28] and decision-level fusion (late fusion) [29]. Early fusion involves concatenating acoustic and textual features into a single vector prior to processing, whereas late fusion trains separate unimodal models and aggregates their independent predictions via voting or averaging mechanisms [9]. Beyond these traditional methods, more advanced techniques employ attention-based fusion; for

example, Tsai et al. [30] utilize a Transformer encoder to integrate signals via cross-modal attention mechanisms, allowing the model to weigh the interaction between modalities dynamically.

2.4 LLMs for Emotion Recognition

The rapid advancement of Large Language Models (LLMs) has revolutionized the field of Affective Computing, particularly within the sub-domain of Emotion Recognition in Conversation (ERC). Recent literature demonstrates that LLMs possess an inherent capability to perceive and reason about human emotional states [31]. Unlike traditional statistical models, LLMs process textual input through complex cognitive-like mechanisms, allowing them to interpret semantic nuances akin to human reasoning [32].

This reasoning capability has been leveraged in various capacities. For instance, Dutta and Ganapathy [33] utilized LLMs as automated annotators to generate pseudo-labels for unlabeled ASR transcripts, successfully supervising the training of smaller, resource-constrained TER models.

Building on this, recent works have shifted the paradigm from discriminative classification to generative emotion recognition. InstructERC [34] was the pioneering framework to formulate ERC as a generative instruction-following task, demonstrating that LLMs could outperform specialized discriminative models by directly generating emotion labels as text. Furthermore, this generative approach has expanded into the multimodal domain. DialogueLLM [35] integrated visual modalities by converting video frames into textual descriptions. By feeding these descriptions into the prompt, they achieved multimodal inference within a purely text-based architecture—a strategy that significantly influences the methodology of this project.

2.5 Speech Incorporation in LLMs

Integrating speech features into LLMs remains an active area of research, with methodologies generally falling into prompt-based or architectural adaptation strategies. SpeechCueLLM

[36], which serves as the foundational baseline for this work, proposes the discretization of numeric acoustic features into natural language ”bins”. These descriptive bins are provided as part of the text prompt alongside the audio transcription, enabling the LLM to infer emotion tags without requiring architectural modification.

A comparable approach was proposed by Santoso et al. [37], who similarly incorporated speech feature descriptors as natural language text within the prompt. However, they extended this methodology by utilizing the LLM-generated labels to supervise the training of a smaller, distilled model. Their results demonstrated that these weakly generated labels could achieve performance comparable to human annotation at a significantly lower cost.

Alternative strategies focus on architectural alignment rather than prompting. SECap [38] employs a HuBERT audio encoder to encode raw audio, utilizing a bridging network to project these vector representations into a latent space compatible with LLaMA. This allows the model to directly decode and generate speech emotion captions. Similarly, Zhang et al. [39] introduced the use of acoustic landmarks as specialized input tokens, enabling the LLM to detect depression directly from spectral patterns.

2.6 Instruction-Tuned Foundation Audio-LLMs

Beyond specialized SER architectures, a substantial body of research focuses on Instruction-Tuned Multimodal LLMs trained across diverse audio tasks, including emotion recognition. These foundation models leverage shared representations across varying auditory domains, often yielding competitive performance on downstream tasks. Prominent examples include **SALMONN** [40], which specializes in general audio perception and paralinguistic analysis. Furthermore, the **Qwen-Audio** series—specifically Qwen2-Audio [41] and the recently released Qwen3-Omni [42]—extends these capabilities into the conversational domain, facilitating end-to-end speech interaction and complex affective reasoning.

Chapter 3

Methodology

This chapter details the baseline methodology, identifies its limitations and sub-optimal configurations, and presents the proposed framework designed to address these challenges.

3.1 Baseline Preliminary

SpeechCueLLM [36] serves as the foundational baseline for this research. Its core methodology involves converting numerical speech features into discretized natural language descriptions (bins), thereby leveraging the inherent Natural Language Understanding (NLU) capabilities of Large Language Models (LLMs) to perform emotion recognition solely from textual input. Crucially, this approach enables the processing of multimodal data through prompt engineering, eliminating the need for complex architectural modifications to the LLM.

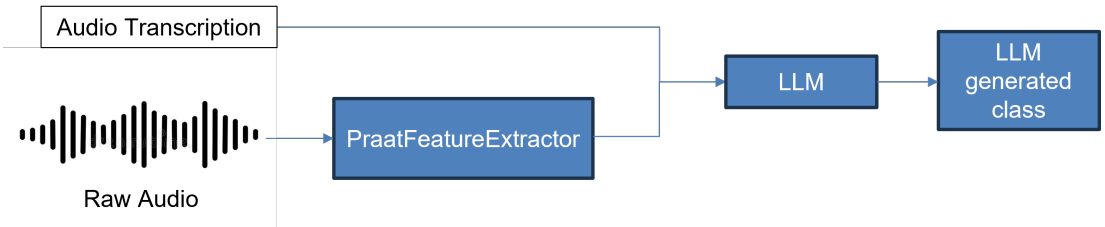

Figure 3.1: SpeechCueLLM Baseline Pipeline

Figure 3.1 illustrates the pipeline. The system utilizes Praat [43] to extract acoustic

features, which are subsequently integrated into the LLM prompt alongside the corresponding audio transcriptions. The model then generates the a single emotional label from its predefined taxonomy as a text output.

3.1.1 Feature Extraction and Discretization

Praat is a widely used, open-source software tool to extract speech features such as pitch and intensity. The baseline system utilise praat to extract five core acoustic features to characterize the prosodic nature of the audio:

- **Average Volume:** Represents the overall loudness or intensity of the speech.
- **Volume Variation:** Captures the dynamic range and fluctuations in intensity.
- **Average Pitch:** Indicates the general fundamental frequency (F_0) of the speaker's voice.
- **Pitch Variation:** Measures the modulation or variability in pitch (standard deviation).
- **Speaking Rate:** Quantifies the speed of speech (syllables per second excluding pauses).

To bridge the modality gap between continuous numerical data and the text-based LLM, the system employs quantile-based discretization. Specifically, continuous values are mapped into five distinct quintiles, corresponding to the natural language categories: *Very Low, Low, Medium, High,* and *Very High*.

Based on these categorical values, the system generates descriptive natural language phrases tailored for LLM consumption. An example of such a speech description is illustrated in Figure 3.2.

3.1.2 Prompt Construction

To fully utilize the In-Context Learning (ICL) capabilities of LLMs, the baseline incorporates a sliding window mechanism with a context width of $N = 8$. This captures the preceding eight conversational utterances as historical context, which are processed

Captions:
No, they told me-I'm sorry, but they told me to be in this line.

Perceptual Analysis:
The speaker conveys a distinct sense of restrained frustration and defensive assertion. The delivery begins with a sharp, impulsive spike on the initial denial (No, they told me...")—suggesting immediate annoyance—before quickly settling into a more controlled, socially acceptable tone during the apology (I'm sorry..."). The voice sounds firm and insistent rather than submissive, yet it stops short of shouting or overt aggression.

Description:
Target speech characteristics: moderate pitch with high variation, high volume with moderate variation, very low speaking rate.

Figure 3.2: SpeechCueLLM Description Example

alongside the target utterance (total window size of 9) to inform the decision-making process. Additionally, pitch features are explicitly appended to the end of the three most recent utterances within the prompt. Empirical analysis by the baseline authors suggests that pitch dynamics in the immediate conversational history yield the highest correlation with accurate emotion prediction. (so we follow the baseline setting)

An example of the constructed prompt is presented in Figure 3.3.

Training Configuration

The baseline model is fine-tuned using Low-Rank Adaptation (LoRA) on the IEMOCAP dataset.

3.2 Limitations of the Baseline

While SpeechCueLLM establishes a strong foundation for LLM-based SER, several limitations hinder its direct applicability to real-world deployment:

Inference-Time Dependencies: A critical limitation is the baseline's reliance on pre-provided ground-truth data, specifically manual transcriptions and gender labels,

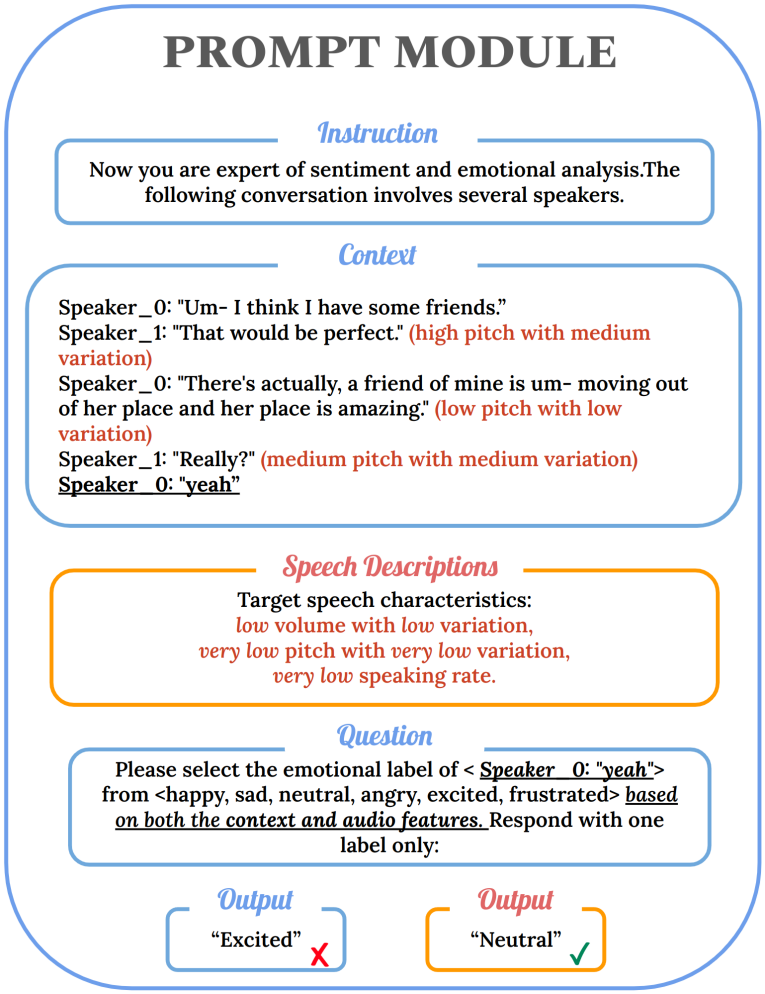

Figure 3.3: Baseline Prompt

alongside the raw audio. This dependence highlights a gap between controlled research conditions and practical deployment, where such metadata is rarely available in real-time.

Restricted Emotion Taxonomy: The baseline is constrained to recognizing only six emotions. This taxonomy fails to encompass the full spectrum of primary emotions, nor does it account for fine-grained secondary states (e.g., *Disappointed, Concerned*), as humans are capable of understanding and recognising more than six emotions **??**.

Unexplored Single-Utterance Performance: While the utilization of historical context via ICL is a key strength, the baseline's efficacy in single-utterance scenarios remains underexplored. Evaluating performance without conversational history is crucial for applications where long-form context is unavailable.

Feature Selection and Prompt Efficiency: The baseline's reliance on a limited set of five acoustic features captures only macro-level dynamics, failing to distinguish emotions with similar prosodic envelopes. For instance, *Anger* and *Excitement* both exhibit high pitch and intensity, creating a "semantic bottleneck" where they become mathematically indistinguishable without finer spectral analysis. While linguistic context offers some disambiguation, it remains insufficient due to the inherent ambiguity of natural text. However, naively expanding the feature set risks "prompt bloat," where excessive token consumption dilutes the LLM's attention mechanism and degrades reasoning efficacy.

Prompt Structure Optimization: While the baseline authors explored adding more descriptive features, they overlooked a fundamental aspect of prompt engineering: the organization of information. The comparison between unstructured natural language descriptions and structured representations remains an open research question.

3.3 Proposed Improvement

The baseline framework presents a computationally efficient foundation by avoiding complex architectural modifications to the LLM. Its method of incorporating speech features via natural language is highly scalable and offers inherent interpretability, as the model conditions its predictions on human-readable descriptions. Building upon these strengths, this section details the proposed enhancements designed to address the identified limitations.

3.3.1 System Overview

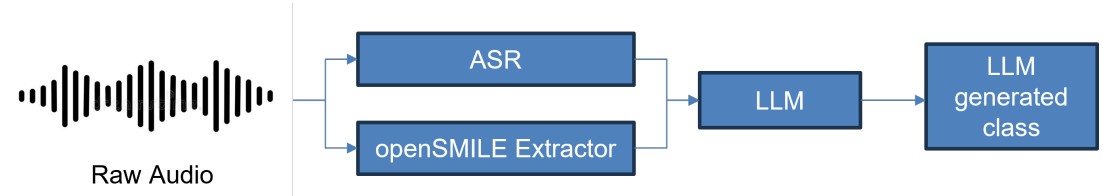

Figure 3.4: SpeechCueLLM Full Pipeline

Figure 3.4 illustrates the revised architecture. Unlike the baseline, this framework op-

erates as a fully automated pipeline requiring only raw audio input, thereby eliminating the dependency on manual transcriptions and metadata during inference.

The system comprises four primary components:

1. **ASR Module:** Automatically generates transcriptions from raw audio in real-time.
2. **Extended Feature Extractor:** Extracts a comprehensive acoustic feature set beyond standard prosody, incorporating spectral descriptors (e.g., MFCCs) and voice quality metrics (e.g., Jitter). These signals capture vocal *timbre* and *physiological tension*, resolving ambiguities in high-arousal states (e.g., Anger vs. Excitement) where pitch and volume are insufficient.
3. **Gender Classifier:** A lightweight logistic regression classifier trained on the extracted features to automatically infer speaker gender.
4. **LLM Classifier:** The central inference engine that integrates these multimodal signals to generate the final emotion tag.

3.3.2 Speech Features Incorporation

eGeMAPS

Basic prosodic descriptors such as pitch, loudness, and speaking rate are insufficient to encompass the full spectral and temporal dimensionality of complex emotional speech. To address this limitation, the proposed framework adopts the **eGeMAPS** (Geneva Minimalistic Acoustic Parameter Set) [44] feature set, extracted using **openSMILE** toolkit [45]. Unlike the baseline, which relies on a limited set of heuristic descriptors, eGeMAPS extracts a standardized vector of 88 acoustic parameters explicitly optimized for paralinguistic analysis.

A critical distinction between the baseline extraction methodology (Praat) and our Proposed Extended Feature Set (eGeMAPS) lies in the depth and perceptual relevance of the acoustic analysis:

- **Perceptual Pitch Scaling:** While Praat typically computes Fundamental Fre-

quency (F_0) on a linear Hertz scale, eGeMAPS transforms pitch into a logarithmic semitone scale. This aligns more closely with human auditory perception, where relative melodic contours are more significant for emotion perception than absolute frequency values.

- **Voice Quality & Timbre:** Beyond prosody, eGeMAPS includes micro-prosodic features such as Jitter (frequency perturbation) and Shimmer (amplitude perturbation), which quantify voice "roughness" and "breathiness." It also incorporates spectral descriptors like Spectral Flux, which captures the rate of change in the power spectrum. These features are indispensable for distinguishing high-arousal negative emotions (e.g., the "harshness" of Anger) from low-arousal ones (e.g., the "breathiness" of Sadness), a nuance often lost in basic prosodic analysis.
- **Spectral Envelope & Vocal Tract Dynamics:** eGeMAPS extends the analysis to the frequency domain via **Mel-Frequency Cepstral Coefficients (MFCCs)** and formant frequencies (e.g., F_1, F_2). Unlike prosodic features, which describe *how* speech is modulated over time, spectral features capture the *resonant properties* of the vocal tract influenced by articulatory configuration. Since distinct affective states often induce subtle changes in vocal tract shape (e.g., smiling vs. frowning), these spectral descriptors enable the model to disentangle acoustically similar yet semantically distinct emotions that prosody alone cannot separate.

However, directly ingesting all 88 eGeMAPS features into the LLM would result in excessive token consumption and potential noise. To resolve this, we employ a hybrid feature selection strategy to isolate the top $K = 20$ most discriminative attributes. This process integrates **Lasso (L1) Regression** [46] to induce sparsity and select features with high predictive value, alongside **ANOVA** [47, 48] to verify statistical significance between emotion classes.

These methods are chosen for their complementary strengths:

- **ANOVA (Filter Method):** Effectively identifies features that possess strong individual correlations with the target emotion, though it may overlook feature redundancy.

- **Lasso Regression (Embedded Method):** Assesses the joint contribution of variables, potentially retaining features that are weak individually but critical when combined (interaction effects).

The selected features from both methods are combined to ensure a comprehensive representation. This data-driven approach ensures flexibility, allowing the system to dynamically adapt the feature set based on the specific acoustic characteristics of different datasets.

3.3.3 Dynamic Feature Masking (DFM)

To enhance model robustness and prevent overfitting to specific acoustic modalities, we implement a Dynamic Feature Masking (DFM) strategy. During training, each acoustic feature token is subject to a masking probability p, which increases linearly per epoch until it reaches a predefined maximum threshold p_{max}. The probability for a given epoch is calculated as:

$$p = p_{max} \times \frac{\text{Current Epoch}}{\text{Total Epochs}}$$

This progressive regularization forces the model to perform inference with increasingly partial information, thereby mitigating over-reliance on any single dominant feature and improving adaptability to noisy, real-world inputs.

3.3.4 Prompt Optimization

The baseline methodology relies on generating natural language narratives to summarize speech features. While intuitive, this approach suffers from poor scalability: as the number of extracted features increases, the length of the descriptive narrative scales linearly. This introduces significant syntactic overhead—an abundance of filler words and connectives that dilutes the information density and distracts the model's attention mechanism from critical acoustic data [49].

To resolve this, the proposed framework transitions from narrative descriptions to a Structured Prompt format (e.g., Key-Value pairs or JSON-like structures).

Figure 3.5: SpeechCueLLM Structured Prompt Example

Description (Omitted some features):
Target speech characteristics:

--- Pitch (Intonation) ---
Average Pitch: Medium
Pitch Stability (StdDev): Medium

--- Loudness (Energy) ---
Average Loudness: Medium
Loudness Variation (StdDev): Low

--- Rhythm & Speed ---
Speaking Rate: Very high

--- Voice Quality (Roughness/Breathiness) ---
Jitter (Voice Roughness): Medium
Shimmer (Voice Breathiness): High

--- Spectral Balance Related Features---
…
--- Formants Related Features---
…
--- MFCCs Related Features---
…

As illustrated in Figure 3.5, this format strips away linguistic redundancy, presenting the acoustic features in a raw, dense representation. Recent literature demonstrates that LLMs possess a robust capability to parse and reason over such structured data [50]. By organizing complex acoustic features into a concise, dictionary-like structure, we maximize information density and significantly reduce token consumption. This approach has been shown to yield performance comparable to, or exceeding, textual descriptions for complex tabular reasoning tasks while maintaining computational efficiency [51].

Recent research demonstrates that LLMs possess a robust capability to parse and reason over structured data [50]. By organizing complex acoustic features into a concise, dictionary-like structure, we maximize information density and reduce token consumption, which has been shown to yield comparable performance for complex data tasks [51].

Furthermore, we implement a **Dynamic Prompt Construction** mechanism to accommodate dataset-specific feature sets. Since the optimal acoustic descriptors are determined empirically for each corpus (e.g., via LASSO or ANOVA feature selection), a rigid template is unsuitable. Our framework dynamically generates the prompt schema at runtime, populating the structured input fields with only the specific feature keys relevant to the target dataset. This process includes a semantic mapping layer that translates raw OpenSMILE variable names into natural language descriptors (e.g., mapping `F0semitoneFrom27.5Hz_sma3nz_amean` to ”Average Pitch”). This step is

essential to bridge the modality gap, ensuring that the technical acoustic features align with the LLM's pre-trained semantic understanding of vocal characteristics.

3.3.5 Dataset Scalability and Generalization

To overcome the limited taxonomy of the IEMOCAP-trained baseline (six emotions), we train two distinct model checkpoints to address different scopes of SER:

1. **Dyadic Model:** Optimized for dyadic interactions, excelling in scenarios requiring historical conversational context.
2. **Utterance-Level Fine-Grained Model:** Optimized for naturalistic, single-utterance scenarios, capable of identifying a broad spectrum of up to 16 fine-grained emotion classes (e.g., *Concerned, Disappointed*) often encountered in the wild.

Chapter 4

Experiments

4.1 Datasets

4.1.1 IEMOCAP

The Interactive Emotional Dyadic Motion Capture (IEMOCAP) corpus [15], collected by the Speech Analysis and Interpretation Laboratory (SAIL) at USC, stands as a premier benchmark for SER. The dataset comprises approximately 12 hours of audiovisual data featuring ten professional actors in five dyadic sessions, performing both scripted and improvised scenarios. While the corpus includes motion capture data, this research utilizes strictly the acoustic and textual modalities.

The dataset is distinguished by its high signal-to-noise ratio, resulting from controlled studio recording conditions. Furthermore, its dyadic structure provides rich conversational history, enabling models to leverage temporal context across dialogue turns—a critical factor for the proposed context-aware framework.

Data Partitioning

Session	1	2	3	4	5
Count	1365	1348	1533	1512	1622

Table 4.1: Utterance Distribution per Session in IEMOCAP

To ensure rigorous evaluation, we adhere to a speaker-independent protocol, isolating Session 5 as the held-out test set while using Sessions 1–4 for training. This strategy ensures that no speaker identity overlaps between training and testing, thereby preventing the model from overfitting to specific speaker characteristics. Table 4.1 confirms that the data is evenly distributed across sessions.

Class Selection and Preprocessing

Emotion	Count	Total Duration (min)
fru	1849	145.16
neu	1708	111.08
ang	1103	82.96
sad	1084	99.30
exc	1041	82.95
hap	595	43.05
sur	107	5.54
fea	40	1.82
oth	3	0.25
dis	2	0.08

Table 4.2: Emotion Distribution of IEMOCAP

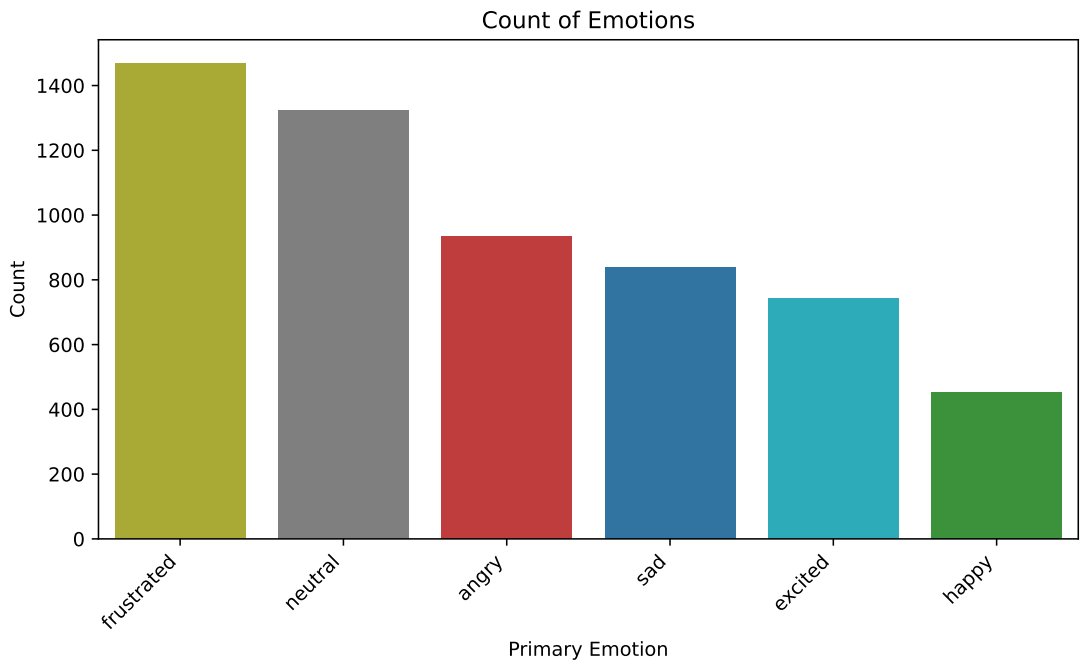

Figure 4.1: IEMOCAP Emotion Count

As shown in Table 4.2 and Figure 4.1, the raw dataset exhibits significant class imbal-

ance. The classes for *Fear* and *Disgust* are severely underrepresented. Thus, we filter the dataset to focus on the four primary emotions plus *Excited* and *Frustrated*, resulting in a final six-class taxonomy: **Happy, Sad, Angry, Frustrated, Excited, and Neutral**.

Statistical Characteristics

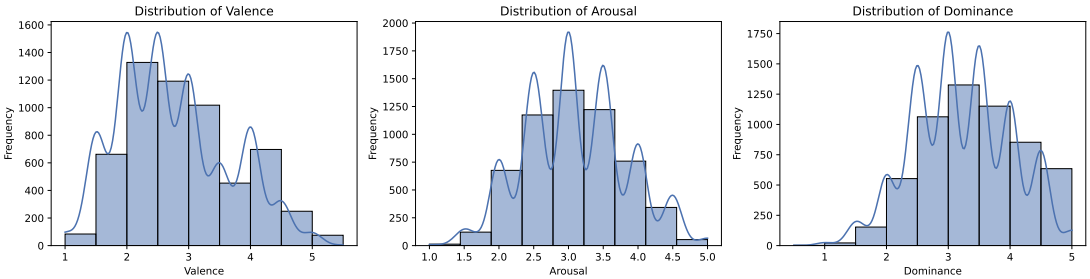

Figure 4.2: VAD Distribution of IEMOCAP

We further analyze the dimensional characteristics of the dataset via VAD scores:

- **Distributions:** Valence exhibits a slight right skew, Arousal is largely symmetric, and Dominance shows a left skew. The mean values are 2.74, 3.09, and 3.21, respectively (Figure 4.2).
- **Demographics:** The dataset maintains a balanced gender ratio of 53% Male to 47% Female.
- **Length:** The average utterance duration is 4.60 seconds.

Valence-Arousal Space Analysis

To investigate the dimensional relationships between emotion classes, we visualize the interaction between Valence and Arousal in a 2D space, grounded in Russell's Circumplex Model [52].

Upon normalizing the IEMOCAP ratings to a scale of $[-1, 1]$ (centered at 0), distinct clustering patterns emerge as illustrated in Figure 4.3. The Valence dimension acts as the primary discriminator, clearly separating positive states (*Happiness, Excitement*) which cluster in the right hemisphere, from negative states (*Anger, Frustration, Sadness*) in the left.

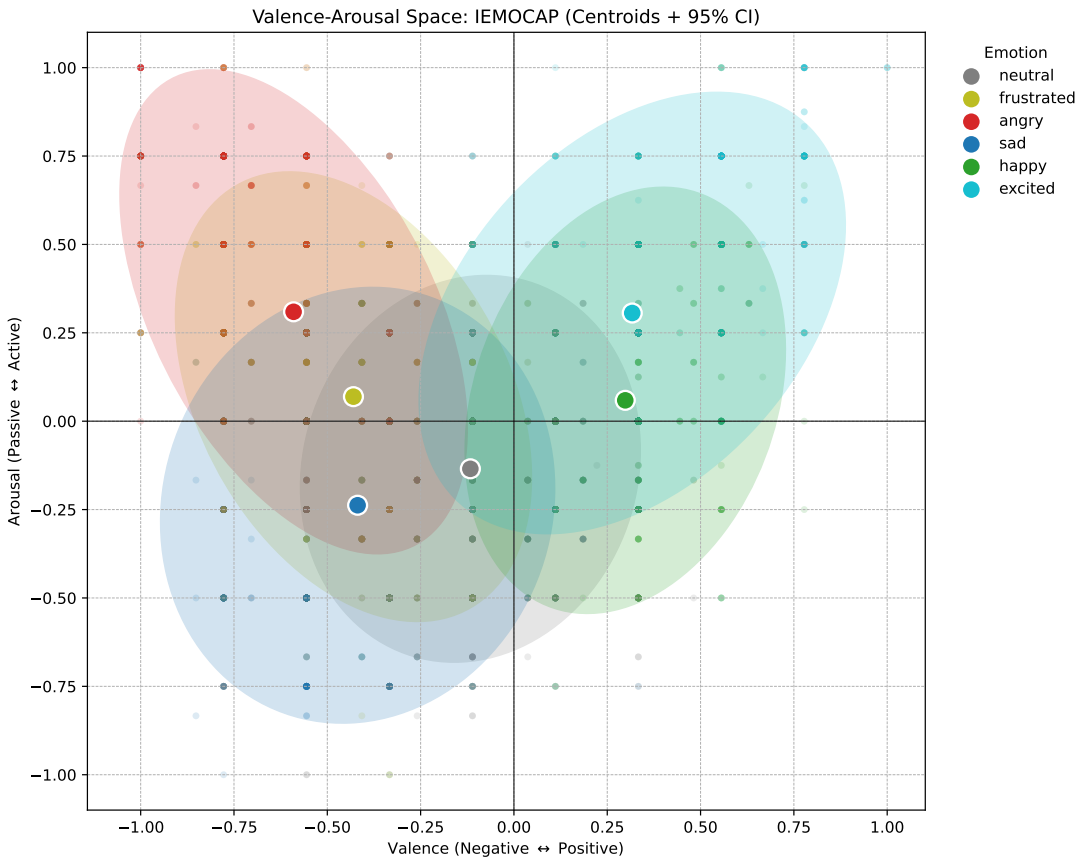

Figure 4.3: Valence-Arousal Space of IEMOCAP (Normalized Centroids)

The Arousal dimension facilitates distinction within these valence groups. On the positive spectrum, *Excitement* exhibits consistently high activation (top-right quadrant), whereas *Happiness* spans a wider range of arousal levels. Conversely, on the negative spectrum, *Sadness* is characterized by low arousal (bottom-left quadrant), sharply contrasting with *Anger*, which occupies the high-arousal region. Notably, *Frustration* mirrors the distributional variance of *Happiness*, spanning a broad range of arousal despite its negative valence. Finally, *Neutral* samples cluster near the origin of the valence axis but skew slightly toward lower arousal values.

4.1.2 MSP

The **MSP-Podcast** corpus [16] represents a significant milestone in naturalistic SER research, culminating from over a decade of continuous data collection. The dataset comprises approximately 409 hours of audio sourced from diverse podcasts and audio-

sharing platforms, featuring a vast and heterogeneous set of speakers. Unlike the acted dyadic interactions in IEMOCAP, MSP-Podcast captures spontaneous, real-world conversations.

Consequently, the emotional expressions within the corpus are often subtle and non-prototypical, contrasting sharply with the exaggerated portrayals found in scripted datasets. The recordings inherently contain varying levels of background noise and acoustic overlapping, presenting a more rigorous challenge for acoustic modeling. Each segment is annotated by a minimum of five human raters, providing both categorical labels and dimensional VAD scores on a 7-point scale. This complexity positions MSP-Podcast as an essential benchmark for evaluating model robustness and generalization in ”in-the-wild” deployment scenarios.

Emotion	**Count**	**Total Duration (min)**
neutral	39677	3569.86
concerned	21797	2146.83
happy	21336	2055.11
sad	11817	1123.84
angry	9588	922.57
frustrated	7552	754.05
amused	6823	640.44
annoyed	6572	648.81
disappointed	6213	589.38
excited	4471	439.03
surprise	4040	367.01
contempt	3745	362.37
confused	2159	195.11
disgust	2138	213.77
fear	697	64.38
depressed	505	49.17

Table 4.3: Emotion Distribution of MSP-Podcast

As illustrated in Table 4.3 and Figure 4.4, the dataset exhibits significant class imbalance, with a preponderance of neutral samples.

Statistical Characteristics

Despite the categorical imbalance, the dimensional VAD distribution appears relatively symmetric. We summarize the key dataset statistics as follows:

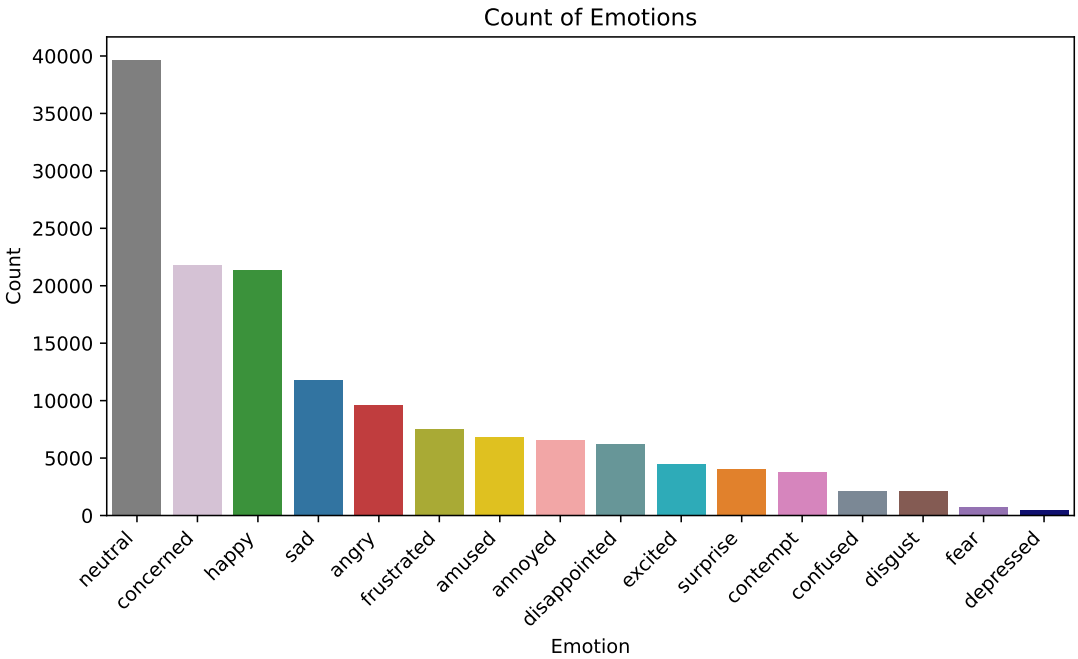

Figure 4.4: MSP-Podcast Emotion Count

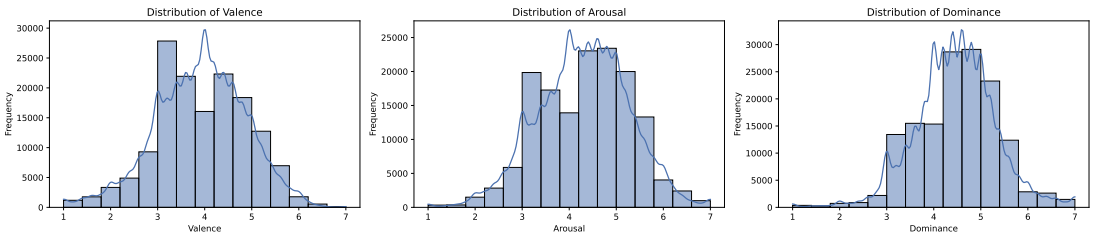

Figure 4.5: VAD Distribution of MSP-Podcast

- **Dimensional Distribution:** The dataset exhibits mean values of 3.91 for Valence, 4.30 for Arousal, and 4.45 for Dominance. While one might initially attribute the symmetry of the distribution (Figure 4.5) to the preponderance of neutral samples, analysis of the balanced subset in Section 4.1.3 demonstrates that this characteristic is intrinsic to the dataset.
- **Demographics:** The dataset maintains a near-balanced gender ratio of 52% Male to 48% Female, minimizing the risk of gender-based bias.
- **Length:** The average utterance duration is 5.69 seconds.

4.1.3 MSP-Podcast Subset

To mitigate the bias introduced by the severe class imbalance observed in the full corpus, we constructed a balanced subset. Specifically, we capped the duration of each emotion class to a maximum of one hour (approximately 500 utterances). This strategy ensures that the model treats all classes with equal importance during optimization, preventing overfitting to majority classes like *Neutral* or *Concerned*.

We then split the resulting balanced dataset into training and test sets using a test ratio of 0.2. The split was performed via stratified subsampling based on emotion labels, ensuring that the test set preserves the same balanced class distribution as the training set.

Emotion	**Count**	**Total Duration (min)**
confused	538	59.66
surprise	529	60.06
neutral	527	59.99
fear	521	60.01
disappointed	513	60.01
happy	510	59.91
angry	509	60.16
amused	504	59.84
contempt	500	60.01
concerned	494	60.01
sad	493	60.01
excited	490	60.00
disgust	484	59.75
frustrated	477	59.81
annoyed	473	60.04
depressed	404	49.17

Table 4.4: Emotion Distribution of the MSP-Podcast Subset

As detailed in Table 4.4, this process yields a significantly more uniform distribution, with most categories containing between 450 and 550 samples. Notably, naturally rare classes such as *Depressed* remain slightly smaller due to data scarcity, but the disparity is drastically reduced compared to the original distribution.

Statistical Validation

To verify that this subsampling preserves the fundamental acoustic characteristics of the corpus, we compared the dimensional distributions of the subset against the full dataset.

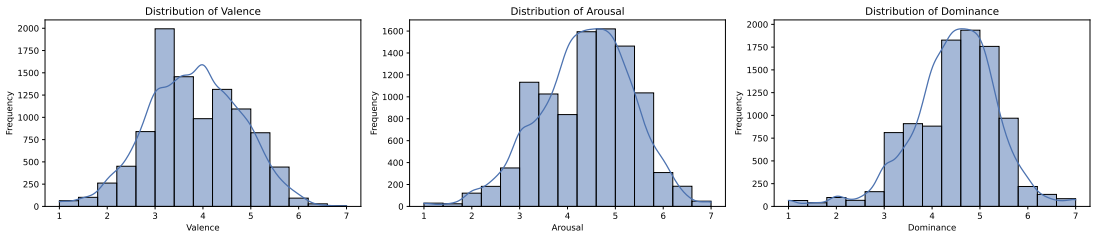

Figure 4.6: VAD Distribution of MSP-Podcast Subset

- **Dimensional Consistency:** As shown in Figure 4.6, the subset preserves the original VAD distribution with remarkable fidelity. The average values for the subset are Valence: 3.82, Arousal: 4.37, and Dominance: 4.47. The fact that the distribution remains centered—even when "extreme" emotions are up-sampled to match "neutral" ones—supports the hypothesis that naturalistic emotional speech is less exaggerated and spectrally closer to the neutral position than acted speech.
- **Demographic Stability:** The subset maintains a gender ratio of 51% Male to 49% Female, preserving the demographic balance of the original corpus.
- **Attribute Preservation:** Table 4.5 compares the mean valence per emotion between the full dataset and the subset. The average deviation is negligible (≈ 0.06) for the majority of classes, indicating that the subset is statistically representative of the broader population.

For the remainder of this report, all references to "MSP-Podcast" denote this balanced subset unless explicitly stated otherwise.

Valence-Arousal Space Analysis

We apply the same dimensional analysis to the MSP-Podcast subset, normalizing the original 7-point scale to a range of $[-1, 1]$. To facilitate a direct and meaningful comparison with IEMOCAP—and to maintain visual clarity—we restrict this visualization

Emotion	Mean (Original)	Mean (Subset)
angry	2.8483	2.7925
annoyed	3.1819	3.1295
frustrated	3.1943	3.1879
disgust	3.3086	3.3025
disappointed	3.3561	3.3571
sad	3.3907	3.3894
contempt	3.5978	3.5615
fear	3.6575	3.6506
concerned	3.7953	3.7349
confused	3.9176	3.9139
neutral	3.9575	3.9708
depressed	3.9642	3.9663
surprise	4.4870	4.4965
amused	4.7855	4.7748
happy	4.8361	4.8269
excited	4.8991	4.9083

Table 4.5: Comparison of Mean Valence between Original Dataset and Subset

to the standard primary emotions (*Happy, Sad, Angry, Fear, Disgust, Surprise, Neutral*) alongside the two secondary emotions present in the IEMOCAP taxonomy (*Excited, Frustrated*).

As illustrated in Figure 4.7, the overall distribution exhibits structural similarities to that of IEMOCAP (Figure 4.3), confirming the universality of the circumplex model. However, specific nuances emerge in the naturalistic domain:

- **Negative Valence Clustering:** Both *Disgust* and *Fear* reside in the negative valence region, consistent with theoretical expectations.
- **Arousal Distinctions:** A divergence is observed in arousal levels. *Disgust* tends toward active arousal, whereas *Fear* exhibits surprisingly passive arousal in this dataset. In fact, the centroid for *Fear* is located in close proximity to the origin, making it the emotion nearest to *Neutral* in terms of Euclidean distance within the VA space. This likely reflects the subtle nature of ”fear” in podcasts (e.g., worry/concern) rather than acute panic.

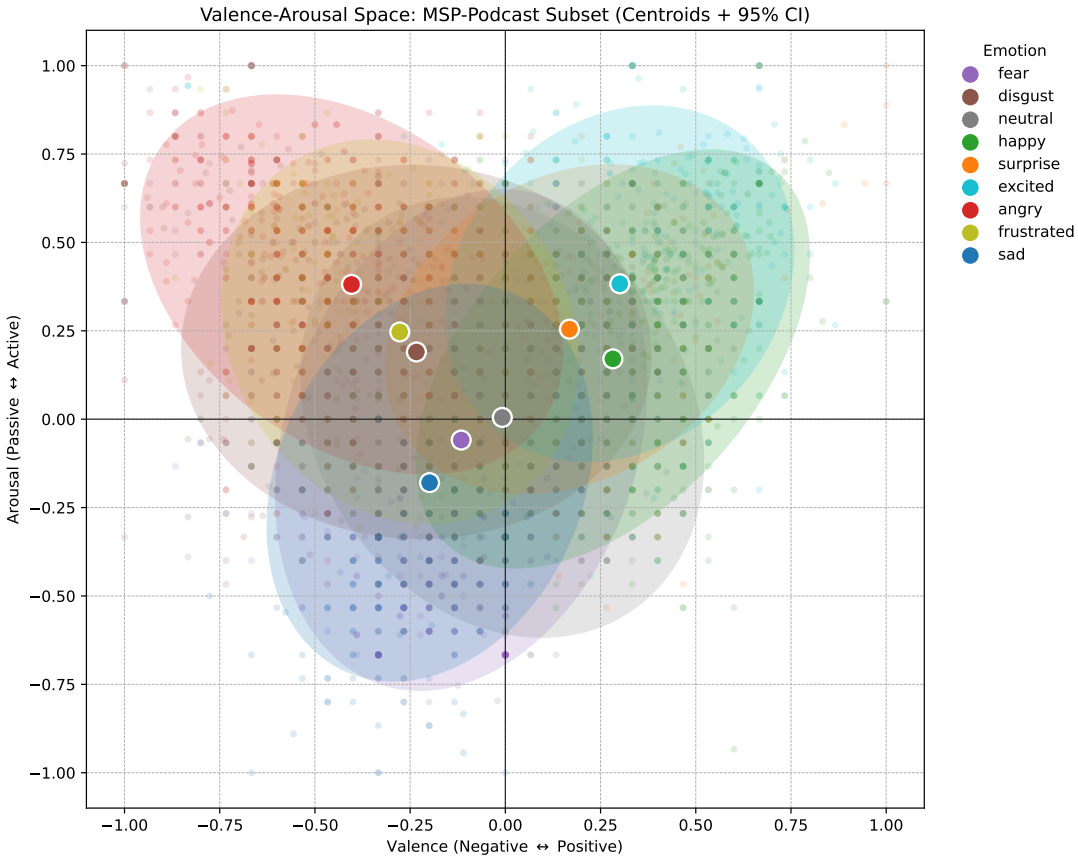

Figure 4.7: Valence-Arousal Space of MSP-Podcast Subset

4.1.4 Datasets Comparison

To bridge the domain gap between acted and naturalistic speech, we perform a comparative analysis of the emotional centroids from both datasets within the Valence-Arousal (VA) space.

As illustrated in Figure 4.8, distinct distributional shifts are observable between the two domains:

- **Negative Valence Extremity:** In the negative valence spectrum, IEMOCAP exhibits significantly more extreme values compared to MSP-Podcast. This phenomenon suggests that professional actors tend to express negative states (e.g., *Anger, Sadness*) with greater intensity and prototypical clarity than what is typically observed in spontaneous conversation. Conversely, in the positive valence spectrum, both datasets converge, with *Excited* and *Happy* showing

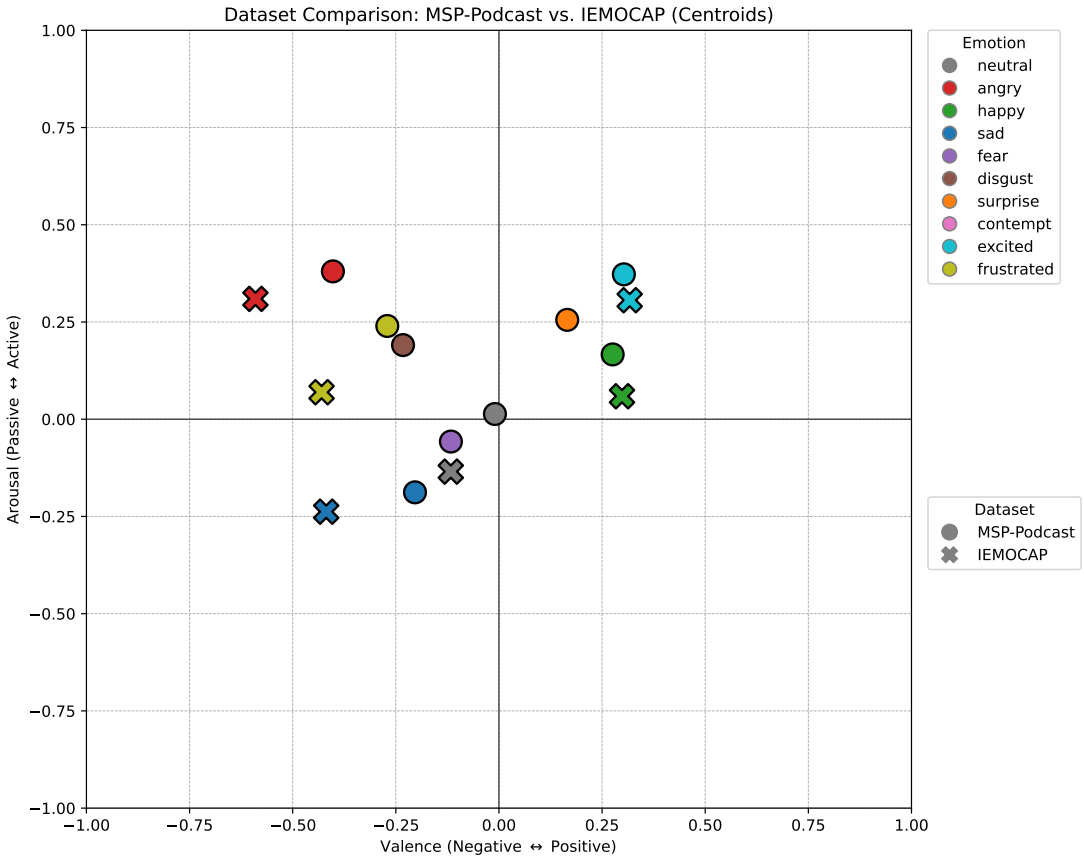

Figure 4.8: Dataset Comparison: IEMOCAP VS MSP-Podcast

comparable average valence.

- **Arousal Baseline Shift:** A systematic shift is observed in the arousal dimension. Contrary to the expectation of ”high-energy” acting, IEMOCAP samples demonstrate a generally lower average arousal compared to their MSP-Podcast counterparts. This may reflect the acoustic reality of podcasts, where background noise, compression, and varying recording conditions artificially inflate energy features, or simply that naturalistic conversations involve more dynamic engagement than controlled studio scripts.

- **Inter-Class Separability vs. Ambiguity:** A critical advantage of IEMOCAP is the wide spatial distribution of its centroids. The clear separation between emotion clusters reduces decision boundary ambiguity, simplifying the classification task. In contrast, MSP-Podcast emotions cluster tightly around the neutral origin. This ”central tendency” reflects the subtle, non-prototypical nature of real-world

emotion, introducing significant ambiguity and complexity to the recognition task.

Class-Wise Dimensional Analysis

To further scrutinize the internal consistency of emotion classes, we analyze the distribution of VA points for each emotion category individually.

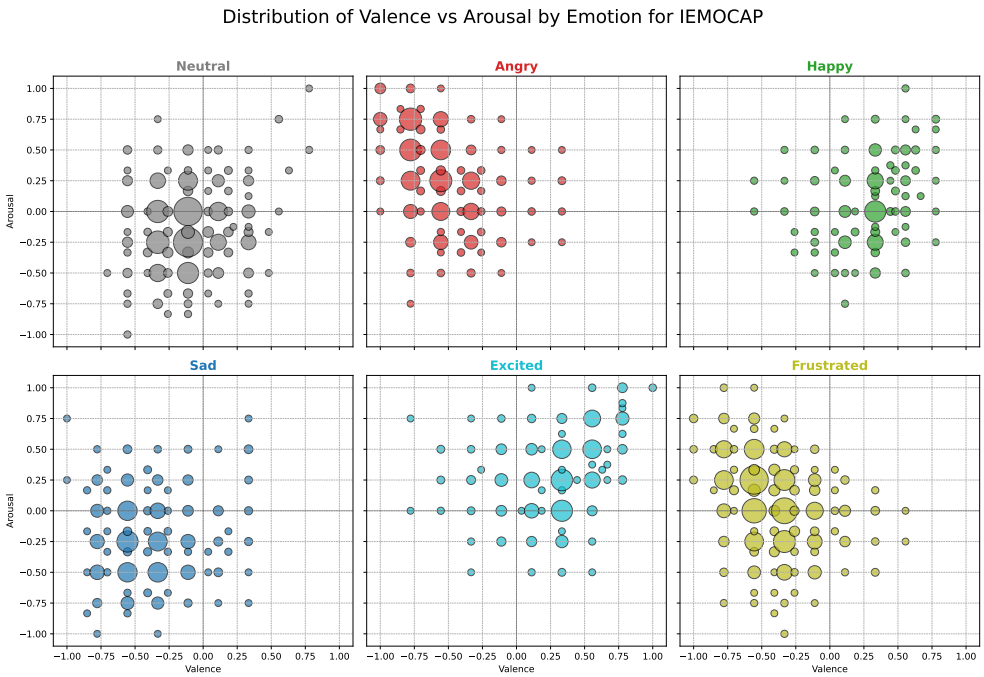

Figure 4.9: Valence-Arousal Space of IEMOCAP by Emotions

Comparative Analysis: These two distributions reveals a fundamental divergence between acted and naturalistic affective spaces.

As observed in Figure 4.9, the IEMOCAP distribution appears highly polarized and discrete. This is most evident in negative emotions such as *Anger*, *Sadness*, and *Frustrated*, where a substantial volume of data points are concentrated beyond the -0.5 valence threshold. This polarization serves as quantitative evidence of theatrical exaggeration; actors in a studio environment tend to project emotions with high magnitude to ensure clarity. While this distinct separation simplifies the classification task, it suggests that the primary challenge in IEMOCAP is not ambiguity, but rather the potential for the model to overfit to these prototypical, high-intensity expressions.

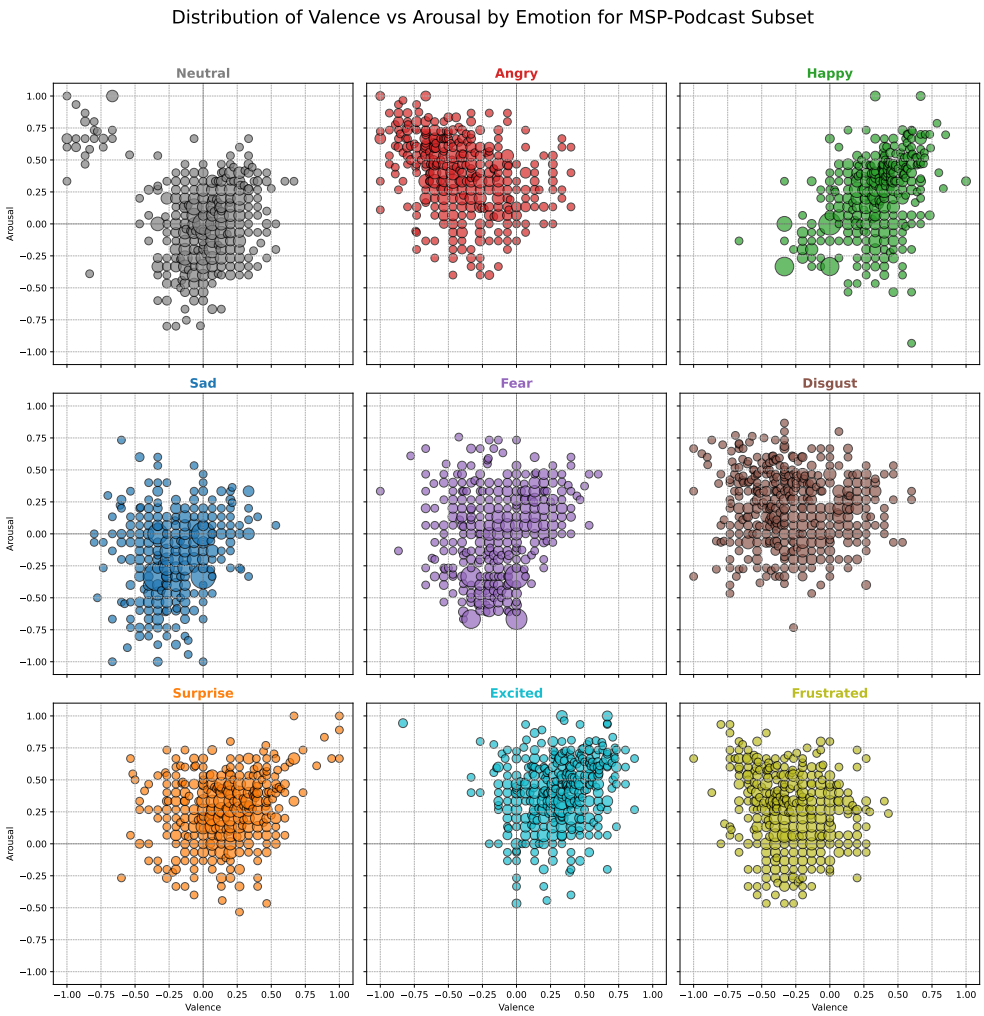

Figure 4.10: Valence-Arousal Space of MSP-Podcast Subset by Emotions

In stark contrast, the dimensional space of the naturalistic MSP-Podcast corpus (Figure 4.10) is markedly more continuous. While the class centroids align with expected quadrants, we observe significant intra-class variance and spectral overlap. Outliers frequently appear in opposing quadrants (e.g., ”Happy” samples with negative valence due to sarcasm or context), creating ”fuzzy” decision boundaries.

The lack of clear separation in MSP-Podcast compared to IEMOCAP introduces irreducible ambiguity, confirming that dimensional coordinates—while powerful—are insufficient as standalone discriminators for naturalistic speech and must be augmented by contextual cues.

4.1.5 Summary of Dataset Analysis

The comparative analysis of IEMOCAP and MSP-Podcast highlights a fundamental trade-off in SER research: Control versus Realism.

- **IEMOCAP (Control):** This dataset provides a controlled environment with high-fidelity, noise-free audio and rich dyadic conversational context. It serves as a standard benchmark for algorithmic validation. However, it is limited by restricted speaker diversity (only 10 actors) and a susceptibility to theatrical exaggeration, potentially yielding models that fail to generalize to subtle, low-arousal cues.
- **MSP-Podcast (Realism):** In contrast, MSP-Podcast captures the complexity of spontaneous human communication, sourced from diverse "in-the-wild" environments. While it accurately reflects naturalistic speech patterns, it introduces significant challenges including varying signal quality, background noise, and acoustic overlapping, alongside high semantic ambiguity in the decision boundaries.

By evaluating on both domains, this research aims to validate that the proposed LLM-based framework can transcend these boundaries—leveraging the clarity of acted speech for learning basic representations while adapting to the noise and nuance of naturalistic environments.

4.2 Training Setup

4.2.1 Implementation Details

The proposed framework is implemented using the PyTorch framework, accelerated via DeepSpeed to optimize memory efficiency on a single Tesla V100-PCIE-32GB GPU.

The preprocessing pipeline integrates several specialized pretrained models:

- **Speech Recognition:** We utilize the **Whisper v3 large-turbo** model [53] for Automatic Speech Recognition (ASR). This model is selected for its optimal

balance between transcription accuracy and computational throughput.

4.2.2 Hyperparameter Configuration

Table 4.6 summarizes the training hyperparameters. Consistent with the baseline, we employ Low-Rank Adaptation (LoRA) [54] for parameter-efficient fine-tuning. By freezing the pretrained LLM weights and injecting trainable rank decomposition matrices, LoRA significantly reduces computational overhead while maintaining model performance.

Hyperparameter	**Dyadic Model (IEMOCAP)**	**Fine-Grained Model (MSP-Podcast)**
LoRA Rank (r)	16	32
LoRA Scaling Factor (α)	16	32
Learning Rate	5×10^{-5}	1×10^{-5}
Optimizer	AdamW	AdamW
Learning Rate Schedule	Constant	Constant
Epochs	15	15
Early Stopping	Yes	Yes
Max Token Length	2500	2500
Max Masking Prob (p_{max})	0.2	0.2

Table 4.6: Comparative Training Hyperparameters for Context-Aware (IEMOCAP) and Fine-Grained (MSP-Podcast) Models

Dataset-Specific Adjustments

LoRA Dimension (r): For the IEMOCAP model, we maintain the baseline rank of $r = 16$. However, for MSP-Podcast, we increase the rank to $r = 32$. This adjustment is necessary because naturalistic, spontaneous speech possesses a higher intrinsic dimensionality and variability compared to acted studio recordings. Increasing the rank provides the adapter modules with greater representational capacity to capture complex, non-linear dependencies in "in-the-wild" emotional expressions.

Learning Rate: We depart from the baseline learning rate of 3×10^{-4}, which empirical observation proved to be too aggressive for our expanded feature set. Consequently, the learning rate for IEMOCAP is scaled down to 5×10^{-5}. For MSP-Podcast, where the optimization landscape is significantly noisier due to naturalistic acoustic variability, the rate is further reduced to 1×10^{-5} to ensure stable convergence.

Optimization & Regularization: Both models are trained for 15 epochs using the **AdamW** optimizer with a constant learning rate schedule. To maximize the utilization of conversational history, the maximum context window is set to 2500 tokens. Additionally, we implement an incremental dynamic feature masking strategy, where the masking probability increases linearly per epoch up to a maximum of $p = 0.2$.

4.3 Evaluation Metrics

To ensure rigorous benchmarking, we employ three primary performance metrics: **Unweighted Accuracy (UA)**, **Weighted Accuracy (WA)**, and **Macro-F1 Score (MF1)**. Let C denote the number of emotion classes, N the total number of samples, and N_i the number of samples in class i.

Weighted Accuracy (WA)

Weighted Accuracy, commonly referred to as standard accuracy, measures the overall ratio of correct predictions across the entire dataset:

$$WA = \frac{\sum_{i=1}^{C} TP_i}{N} \tag{4.1}$$

where TP_i represents the number of True Positives for class i. While intuitive, WA is biased towards majority classes in imbalanced datasets; however, we report it to maintain comparability with baseline studies that utilize this standard.

Unweighted Accuracy (UA)

Unweighted Accuracy (also known as Macro-Recall) is defined as the arithmetic mean of the recall scores per class:

$$UA = \frac{1}{C} \sum_{i=1}^{C} \text{Recall}_i = \frac{1}{C} \sum_{i=1}^{C} \frac{TP_i}{N_i} \tag{4.2}$$

By weighting each class equally regardless of its sample size, UA provides an unbiased estimate of model performance. This metric is particularly critical for evaluating the

MSP-Podcast dataset, ensuring that minority classes (e.g., *Sadness*) are not overshadowed by majority classes (e.g., *Neutral*).

Macro-F1 Score (MF1)

The F1-score is the harmonic mean of precision (P_i) and recall (R_i) for a specific class i, providing a balanced measure that penalizes extreme disparities between the two:

$$F1_i = 2 \cdot \frac{P_i \cdot R_i}{P_i + R_i} \tag{4.3}$$

The Macro-F1 score is subsequently calculated as the unweighted average of these class-wise F1 scores:

$$MF1 = \frac{1}{C} \sum_{i=1}^{C} F1_i \tag{4.4}$$

MF1 serves as a robust metric for multi-class classification, offering a comprehensive assessment of the model's ability to minimize both False Positives and False Negatives across all emotional categories.

4.4 Baseline & SOTA Models

To rigorously assess the efficacy of the proposed framework, we benchmark performance against both the established baseline (SpeechCueLLM) and a diverse set of state-of-the-art (SOTA) models. These comparisons are categorized into two distinct architectural paradigms: Conventional Non-LLM Baselines and Multimodal LLM Baselines.

4.4.1 Conventional Non-LLM Baselines

These models represent the current state-of-the-art in pure audio representation learning. They function by encoding raw audio waveforms into dense vector representations, which are subsequently used for emotion classification without textual or semantic conditioning.

- **WavLM** [55]: A robust large-scale model pre-trained on massive unlabeled

speech data, optimized for diverse downstream speech tasks including SER.

- **emotion2vec** [13]: A specialized model pre-trained specifically for emotion recognition tasks using self-supervised learning on emotional speech data.

4.4.2 Multimodal LLM Baselines

To evaluate the capabilities of general-purpose foundation models in the SER domain, we conduct zero-shot evaluations on several leading LLMs. For these experiments, we replicate the baseline methodology: conversation history is provided as textual context, followed by the target utterance's transcription and its acoustic representation (either as textual features or raw audio, depending on the model's native modality).

- **LLaMA-3-8B-Instruct** [56]: A state-of-the-art **text-only** foundation model.
- **Qwen2-Audio-7B-Instruct** [41]: A Large Audio Language Model (LALM) designed for joint speech and text processing.
- **Qwen3-Omni-30B-A3B-Instruct** [42]: An advanced iteration of the **multi-modal** Qwen series, capable of natively analyzing text, audio, and visual data streams.
- **Qwen3-Omni-30B-A3B-Captioner** [42]: A specialized variant explicitly fine-tuned from *Qwen3-Omni-30B-A3B-Instruct* to generate detailed descriptive captions from audio inputs.
- **DeSTA2.5-Audio** [57]: A general-purpose LALM that integrates *Llama3.1-8B-Instruct* with a *Whisper-large-v3* encoder. It is designed for robust auditory perception and instruction-following tasks, including audio captioning and direct classification.

Methodological Adaptation

While the conversation history and target transcription are consistently provided as text for all models, the **acoustic input strategy** varies by modality:

- **Text-Only Models (LLaMA-3):** We provide the standard structured prompt

containing the **structured feature descriptors** alongside the conversation history.

- **Audio-Capable Models (Qwen/DeSTA):** We provide the conversation history text with the **raw audio path** of the target utterance. This replaces the textual feature descriptions, allowing the model to leverage its internal audio encoder to extract paralinguistic cues directly from the signal while retaining the semantic context of the dialogue.

Furthermore, as *Qwen3-Omni-30B-A3B-Captioner* and *DeSTA2.5-Audio* are capable of both generative description and direct classification, we evaluate both prompting strategies: 1. **Direct Tagging:** Instructing the model to output a specific emotion label. 2. **Caption Generation:** Instructing the model to describe the audio, followed by a keyword filtering process to map the generated description to an emotion category. For each model, we report the higher score obtained between these two strategies to ensure a fair assessment of its maximum capability.

4.5 Ablation Studies

To isolate the distinct contributions of the proposed enhancements, this section systematically evaluates each component of the framework. We analyze the impact of prompt structuring and feature masking across both zero-shot and fine-tuned configurations for the IEMOCAP and MSP-Podcast datasets.

4.5.1 Prompt Structure Analysis

We hypothesize that a structured prompt—presenting speech features in a formalized tabular or key-value format—significantly improves the LLM's ability to parse and attend to acoustic information compared to dense natural language descriptions. To validate this, we compare two prompt configurations:

- $P_{unstruct}$ **(Unstructured):** A naive baseline configuration where transcriptions and acoustic features are concatenated into a single, continuous paragraph without clear headers or delimiters.

- P_{struct} **(Structured):** The proposed methodology, utilizing a structured, table-like format to organize speech characteristics, designed to maximize information density and retrieval efficiency.

4.5.2 Impact of Dynamic Feature Masking

This experiment assesses whether introducing noise via feature masking enhances model robustness and generalization. All trials in this stage utilize the structured prompt (P_{struct}) with the top-20 selected features. We compare two training regimes:

- M_{none} **(No Masking):** A control setting where all acoustic feature tokens are fully visible to the model throughout the training process.
- M_{mask} **(Feature Masking):** The proposed regularization strategy, where each acoustic feature token is subject to a dynamic masking probability during training. This probability increases incrementally per epoch, reaching a maximum of 20%. This mechanism encourages the model to learn redundant feature correlations and reduces over-reliance on any single acoustic parameter.

Chapter 5

Result Analysis

This chapter presents a comprehensive evaluation of the proposed framework. We begin with a comparative performance analysis against established baselines, followed by a granular class-wise examination. Subsequently, we explore cross-dataset generalization capabilities and conclude with ablation studies to isolate the contributions of specific architectural components.

5.1 Overview

This chapter presents a comprehensive evaluation of the proposed framework. We begin with a comparative performance analysis against established baselines on both the IEMOCAP and MSP-Podcast datasets, followed by a granular class-wise examination. Subsequently, we investigate the model's cross-domain generalization capabilities.

To validate our architectural choices, we conduct ablation studies isolating the specific contributions of the structured prompting strategy. Then, we assess its robustness to transcription errors introduced by ASR. Finally, the chapter concludes with a qualitative analysis, illustrating the LLM's step-by-step reasoning process through specific prompt examples and prediction outputs.

5.2 Result Comparison with Baselines

To rigorously assess the proposed framework, we benchmark it against a diverse suite of State-of-the-Art (SOTA) models, ranging from specialized audio-only architectures to large-scale multimodal LLMs.

Performance Analysis

Table 5.1: Main Performance Comparison (%)

	IEMOCAP			MSP-Podcast		
Model Configuration	**UA ↑**	**WA ↑**	**MF1 ↑**	**UA ↑**	**WA ↑**	**MF1 ↑**
Finetuned Non-LLM Baselines						
Emotion2Vec	63.33	67.94	62.26	17.58	17.95	14.45
WavLM	49.0	52.0	48.0	17.0	17.0	13.0
Pretrained LLM Baselines						
Pretrained Llama-3-8B	48.54	46.89	45.52	10.92	11.23	8.43
Qwen2-Audio	45.0	42.0	41.0	12.0	13.0	9.0
Qwen3-Omni	59.0	57.0	54.0	20.0	21.0	19.0
Qwen3-Omni Captioner	–	–	–	16.46	16.96	–
DeSTA-2.5	–	–	–	20.00	20.27	–
Finetuned LLMs						
SpeechCueLLM	70.63	71.97	70.66	18.97	19.26	18.57
Ours	**70.87**	**72.38**	**71.41**	**23.89**	**24.27**	**22.53**
Ours with GT VAD	**74.17**	**75.09**	**73.90**	**31.46**	**31.95**	**31.17**

As detailed in Table 5.1, the proposed framework consistently outperforms the SpeechCueLLM baseline and zero-shot LLM competitors across both domains.

- **MSP-Podcast (Naturalistic):** We observe a substantial performance leap, with Macro-F1 improving by over **3.96%** (absolute) compared to the baseline. This validates the efficacy of our expanded feature set and structured prompting strategy in capturing the nuances of "in-the-wild" speech where linguistic content alone is often ambiguous.

- **IEMOCAP (Acted):** Improvement is present but modest, with a **0.24%** increase in Unweighted Accuracy (UA). This marginal gain is attributed to the dominance of conversational context in dyadic scenarios. In IEMOCAP, the long-term

dialogue history provides strong cues that often overshadow acoustic subtleties, making the model less reliant on the enhanced feature set. Nevertheless, the consistent gain confirms that the proposed mechanism contributes additive value even in context-rich environments.

5.2.1 Class-wise F1 Scores

IEMOCAP Analysis

Table 5.2: Class-wise F1-score analysis on IEMOCAP

Emotion Class	**Baseline (%)**	**Ours (%)**	**Gain (Δ)**
Neutral	**74.2**	73.4	-0.8
Happiness	**60.4**	58.2	-2.2
Sadness	**83.9**	83.2	-0.7
Anger	63.7	**70.7**	+7.0
Excited	70.4	**72.7**	+2.3
Frustrated	**71.3**	70.3	-1.0
WA	*72.0*	***72.4***	*+0.4*

As presented in Table 5.2, the proposed framework exhibits a distinct performance trade-off. While we observe slight regressions in low-arousal and ambiguous categories (e.g., *Neutral* and *Happiness*), these are outweighed by substantial improvements in high-arousal states. Most notably, the recognition of *Anger* sees a dramatic increase of **+7.0%**, while *Excited* improves by **+2.3%**. This suggests that the inclusion of extended acoustic features—particularly spectral dynamics and intensity metrics—is highly effective at resolving the energetic signatures of active emotions, driving an overall improvement in Weighted Accuracy (+0.4%) despite the minor degradation in subtler classes.

MSP-Podcast Analysis

As detailed in Table 5.3, the proposed framework outperforms the baseline, achieving **+3.9%** increase in Macro F1-score. The model demonstrates efficacy in disentangling acoustically distinct secondary emotions, with the largest gains observed in **Confused** (+11.3%), **Disappointed** (+9.8%), and **Excited** (+9.3%). This validates that the inclu-

Table 5.3: Class-wise F1-score analysis on MSP-Podcast (16 classes)

Emotion Class	**Baseline (%)**	**Ours (%)**	**Gain (Δ)**
Primary Emotions			
Neutral	18.5	**21.6**	+3.1
Happiness	15.3	**24.4**	+9.1
Sadness	17.4	**23.4**	+6.0
Anger	18.6	**25.5**	+6.9
Fear	21.4	**25.2**	+3.8
Disgust	26.1	**33.8**	+7.7
Surprised	**23.0**	20.3	-2.7
Contempt	**8.0**	1.5	-6.5
Fine-Grained & Secondary States			
Excited	22.8	**32.1**	+9.3
Amused	20.8	**21.4**	+0.6
Frustrated	17.4	**20.1**	+2.7
Annoyance	14.4	**20.3**	+5.9
Depression	**9.0**	0.0	-9.0
Disappointed	10.6	**20.4**	+9.8
Concerned	20.3	**25.5**	+5.2
Confused	33.5	**44.8**	+11.3
Macro F1-Score	*18.6*	***22.5***	*+3.9*

sion of fine-grained eGeMAPS features (e.g., spectral flux and jitter) provides the necessary resolution to distinguish high-energy states (Excited) from low-arousal negative states (Disappointed), a nuance often lost in the baseline's coarse prosodic modeling.

However, this performance boost entails a specific trade-off regarding extremely subtle or minority classes. We observe a complete collapse in the recognition of **Depression** (-9.0%) and a significant drop in **Contempt** (-6.5%). This suggests that the model effectively optimizes for the majority of distinct emotional archetypes but struggles to isolate these highly ambiguous, low-valence states, likely conflating *Depression* with *Sadness* (which conversely improved by +6.0%) and *Contempt* with *Anger* or *Disgust* due to significant spectral overlap.

5.2.2 Confusion Matrix

5.3 Impact of Emotional Dimensions (Oracle Analysis)

In the preliminary stages of this research, we investigated the integration of emotional dimensions—Valence, Arousal, and Dominance (VAD)—as explicit feature inputs within the structured prompt. Theoretically, these continuous coordinates provide a strong discriminative signal, reducing the high-dimensional search space of categorical emotions (e.g., distinguishing *Excited* from *Happy* based on intensity levels).

To quantify the theoretical upper bound of the framework, we conducted an Oracle Analysis, contrasting the model's performance under two conditions:

1. **Ideal Scenario (Oracle):** The model is conditioned on Ground Truth (GT) VAD scores extracted directly from the dataset labels.
2. **Practical Scenario:** The model is conditioned on predicted VAD scores generated by an external regression model.

5.3.1 The Theoretical Upper Bound (Ground Truth)

Table 5.4 presents the results of the Oracle evaluation. When provided with perfect dimensional information, the Proposed Framework achieves state-of-the-art performance, with Unweighted Accuracy (UA) surging to **74.17%** on IEMOCAP and **31.46%** on MSP-Podcast.

Table 5.4: Oracle Analysis: Impact of Incorporating Ground Truth (GT) VAD Scores. The inclusion of perfect dimensional data yields significant gains over the baseline and the standard proposed mechanism.

	IEMOCAP			**MSP-Podcast**		
Model Configuration	**UA ↑**	**WA ↑**	**MF1 ↑**	**UA ↑**	**WA ↑**	**MF1 ↑**
SpeechCueLLM (Baseline)	70.63	71.97	70.66	18.97	19.26	18.57
Proposed Mechanism (No VAD)	70.87	72.38	71.41	23.89	24.27	22.53
Proposed + GT VAD (Oracle)	**74.17**	**75.09**	**73.90**	**31.46**	**31.95**	**31.17**

These results confirm that VAD coordinates are highly potent indicators of emotional

states. However, this configuration relies on an unrealistic assumption: the availability of perfect affective labels during inference.

5.3.2 VAD Regression Performance (CCC Analysis)

Before evaluating the downstream classification performance using predicted VAD scores, it is crucial to assess the reliability of the upstream regressor itself. We utilise the pretrained regression model from [58] to predict continuous VAD scores. We utilize the Concordance Correlation Coefficient (CCC) as the primary metric, which measures the agreement between the predicted and ground truth continuous traces (where 1.0 indicates perfect alignment).

Table 5.5: Concordance Correlation Coefficient (CCC) of the VAD Regressor. Lower CCC scores indicate higher noise in the predicted primitives, particularly in the Valence dimension for IEMOCAP.

Dataset	**Valence**	**Arousal**	**Dominance**
IEMOCAP	0.5033	0.6775	0.5014
MSP-Podcast	0.5928	0.6558	0.5360

As evident in Table 5.5, the regression performance exhibits a distinct disparity across affective dimensions.

- **Arousal is robust:** Arousal achieves the highest CCC across both datasets ($\approx 0.66 - 0.67$), likely because acoustic energy and pitch variation are strong, linear indicators of activation that are easily captured by the regression model.
- **Valence is the bottleneck:** Crucially, the CCC for Valence is significantly lower, particularly on IEMOCAP (0.5033). Validating the "spectral overlap" hypothesis, the regressor struggles to distinguish positive from negative valence based solely on acoustics.

This mediocre regression performance—specifically the high noise in Valence estimation—foreshadows the challenges in the subsequent classification task.

5.3.3 The Practical Bottleneck (Predicted VAD)

To assess feasibility in a real-world pipeline, we replaced the Ground Truth labels with VAD scores predicted by a pre-trained estimator. As illustrated in Table 5.6, this introduces a critical bottleneck.

Table 5.6: Impact of pred VAD(%)

	IEMOCAP			MSP-Podcast		
Model Configuration	**UA ↑**	**WA ↑**	**MF1 ↑**	**UA ↑**	**WA ↑**	**MF1 ↑**
SpeechCueLLM	70.63	71.97	70.66	18.97	19.26	18.57
Proposed Mechanism with GT VAD	**74.17**	**75.09**	**73.90**	**31.46**	**31.95**	**31.17**
Proposed Mechanism with pred VAD	65.80	68.06	66.40	23.51	23.77	23.47

On the IEMOCAP dataset, the use of predicted VAD scores results in a substantial performance degradation, dropping the UA to 65.80%—significantly lower than the baseline (70.63%). This phenomenon is attributed to error propagation: the LLM learns to rely heavily on the VAD prompts as ”shortcuts” during training (where they are perfect). During inference, when these ”shortcuts” are noisy or inaccurate, the model’s reasoning collapses.

However, the MSP-Podcast dataset exhibits a different behavior. While performance drops relative to the Oracle setting (UA 31.46% → 23.51%), the model with predicted VAD still outperforms the SpeechCueLLM baseline (UA 18.97%). This suggests that for naturalistic data, even noisy dimensional priors provide valuable directional cues that help disentangle complex emotional states, whereas in the cleaner, acted IEMOCAP environment, any noise in the input outweighs the informational gain.

Strategic Exclusion: (For Interim Report Only) Consequently, despite the theoretical promise of dimensional integration, VAD features were excluded from the final Proposed Framework. This decision ensures that the system remains autonomous and robust, deriving its predictions solely from raw acoustic features and text rather than depending on the accuracy of upstream regression components. Future work may revisit this integration as the accuracy of standalone VAD estimators improves.

5.4 Cross-Corpus Generalization Analysis

To evaluate the intrinsic robustness of the learned representations, we conduct a Zero-Shot Domain Transfer assessment. In this protocol, the model is trained on one corpus (Source) and evaluated on the other (Target) without any fine-tuning on the target domain.

Table 5.7: Cross-Dataset Evaluation. To ensure comparability, the MSP-Podcast test set is filtered to the intersection of classes (Anger, Sadness, Happiness, Neutral, Excited, Frustrated). *Note: 'Restricted Decoding' is applied to MSP-trained models to constrain output to the 6-class taxonomy.*

Target Test Set	Training Source	WA (%)	MF1 (%)
MSP-Podcast	IEMOCAP		
	MSP-Podcast		
IEMOCAP	IEMOCAP		
	MSP-Podcast		

Methodology

To ensure a valid comparison despite the differing taxonomies, we impose the following constraints:

- **IEMOCAP → MSP-Podcast:** We apply Target Label Intersection, evaluating only on the subset of MSP samples that correspond to the 6 emotion classes present in IEMOCAP. This tests the model's ability to identify "known" emotions in a novel, naturalistic acoustic environment.
- **MSP-Podcast → IEMOCAP:** We employ Taxonomy Constrained Decoding, where the prompt explicitly restricts the LLM's output space to the 6 IEMOCAP categories. This forces the model to map its diverse 16-class knowledge onto the coarser 6-class target schema.
- **MSP-Podcast → MSP-Podcast (In-Domain Control):** To establish a fair baseline for the cross-corpus performance, we evaluate the MSP-trained model on the same filtered subset used above. Crucially, we apply the same Taxonomy

Constrained Decoding (restricting output to the 6 classes) to isolate the impact of domain shift from differences in label granularity.

Discussion

5.5 Ablation Study

To isolate the contribution of specific architectural choices, we conducted a component-wise ablation analysis. The results, presented in Table 5.8, highlight the incremental gains achieved by each proposed module. Note that the final row represents the fully optimized proposed framework (incorporating Dynamic Feature Masking) for comparative context.

Table 5.8: Ablation Studies: Impact of Prompt Structure and Feature Set on Model Accuracy (%)

Ablation Setting	**FineTuned**	**Dyadic Model (IEMOCAP)**	**Fine-Grained Model (MSP-Podcast)**
Unstructured Baseline Features	×	–	11.23
Structured Baseline Features	×	–	13.09
Structured + eGeMAPS ($K = 20$)	×	–	
Unstructured Baseline Features	✓	–	19.26
Structured Baseline Features	✓	–	
Structured + eGeMAPS ($K = 20$)	✓	–	
Structured + Masked eGeMAPS ($K = 20$)	✓	–	24.27

Analysis

5.6 Impact of ASR Errors

To evaluate the model's resilience to transcription errors, we replace the ground truth text for both the historical conversation and the target utterance with hypotheses generated by Whisper-v3-large-turbo.

Crucially, we utilize the raw ASR output without standard text normalization (e.g., stripping punctuation or lowercasing). We hypothesize that orthographic features such

as exclamation points, ellipses, and capitalization often encode vital paralinguistic cues that contribute to emotional interpretation.

Dataset	Raw Text WER (%)	Normalized Text WER (%)
IEMOCAP	32.27	15.48
MSP-Podcast	34.01	14.06

Table 5.9: Word Error Rate (WER) Analysis of Whisper-v3-large-turbo

WER Analysis

Table 5.9 presents the error rates for the ASR model. We observe a significant disparity between the Raw WER (> 32%) and the Normalized WER (< 15.5%). This delta indicates that approximately 50% of the measured "errors" stem from stylistic divergences—such as punctuation placement or capitalization styles—rather than semantic corruptions. Consequently, the core linguistic content required for emotion inference remains largely intact.

5.6.1 Performance Robustness

To quantify the downstream impact of these ASR errors, we compare the model's performance when conditioned on Ground Truth (GT) versus ASR hypotheses.

		IEMOCAP		MSP-Podcast	
Model	**Text Source**	**UA**	**MF1**	**UA**	**MF1**
Proposed Framework	Ground Truth	**70.86**	**71.41**	23.89	22.53
	ASR Hypothesis	68.10	68.58	**24.11**	**22.88**

Table 5.10: Impact of ASR Error on Model Performance. We compare the robustness of the proposed model when using Ground Truth (GT) transcripts versus ASR hypotheses.

As demonstrated in Table 5.10, the impact of ASR noise varies significantly across domains. On the IEMOCAP dataset, we observe a standard degradation pattern, with Unweighted Accuracy (UA) declining by approximately **2.8%** (70.86% → 68.10%). This drop is consistent with the introduction of transcription errors in a controlled environment where the Ground Truth is highly accurate.

Conversely, the results on MSP-Podcast reveal an intriguing anomaly: the model achieves a slight *performance gain* when conditioned on ASR hypotheses (23.89% → 24.11%). We hypothesize that this stems from the specific segmentation characteristics of the MSP-Podcast corpus, where audio cuts often result in truncated or grammatically incomplete sentences in the Ground Truth. ASR models, in their attempt to transcribe, often perform implicit "semantic completion," effectively guessing the missing context to produce syntactically coherent sentences. In this specific scenario, this "hallucinated" coherence appears to be beneficial, providing the LLM with a more structured linguistic input than the fragmented Ground Truth, thereby facilitating better emotion inference.

5.7 Qualitative Examples

To illustrate the practical impact of the proposed feature integration, we present a case study comparing the decision-making process of the baseline model versus the proposed framework on a challenging sample from the MSP-Podcast test set.

5.7.1 MSP-Podcast Prompt

Context: The target utterance involves a speaker expressing frustration about a legal injustice. The text itself is semantically ambiguous, plausibly supporting an interpretation of either *Anger* or *Disappointment*.

Baseline Prediction (Failure Case)

Baseline Prompt

Prompt Input: Now you are expert of sentiment and emotional analysis. The following conversation noted between '### ###' involves several speakers. The last three utterances are followed by its speech features. ###
Speaker_0:"yeah, it's not fair. i think the saddest thing is that there's law that lets them do this. there's literal law that lets them do this." (medium pitch with high variation) ###

Target speech characteristics: moderate volume with moderate variation, moderate pitch with high variation, moderate speaking rate.
Please select the emotional label of ”yeah, it’s not fair. i think the saddest thing is that there’s law that lets them do this. there’s literal law that lets them do this.” from [angry, frustrated, disgust, annoyed, sad, depressed, disappointed, fear, happy, surprise, excited, contempt, amused, concerned, confused, neutral] based on both the context and audio features. Respond with just one label:

Model Output: `anger`

Ground Truth Label: `disappointed`

Analysis: The baseline model defaults to *Anger*. This is likely driven by the semantic content (”it’s not fair”), which strongly implies injustice—a common trigger for anger. The vague acoustic descriptor (”moderate pitch with high variation”) is insufficient to contradict this textual bias.

Proposed Framework (Success Case)

Proposed Framework Prompt (Condensed)

Prompt Input: *[...Standard Instructions Omitted...]*

Target speech characteristics:

... *[Pitch features omitted]* ...

— Loudness —

Loudness Variation (StdDev): **Low**

... *[Other Loudness features omitted]* ...

— Voice Quality —

Jitter (Voice Roughness): **High**

Shimmer (Voice Breathiness): Medium

Transcript: ”yeah, it’s not fair. i think the saddest thing...”

Please select the emotional label... *[...remainder of prompt identical...]*

Model Output: `disappointed`

Ground Truth Label: `disappointed`

Analysis: The proposed model correctly predicts *Disappointed*. This correction is driven by specific feature cue:

- **High Jitter:** The inclusion of voice quality metrics is decisive here. High jitter indicates vocal instability or roughness, a prosodic feature frequently correlated with sadness, grief, and disappointment.

This comparison illustrates a fundamental advantage of the proposed framework. While the baseline's reliance on broad qualitative descriptors (e.g., "medium pitch", "high variation") captures general acoustic trends, it often lacks the resolution to distinguish between emotionally adjacent states.

By integrating precise, quantitative feature values—such as voice quality metrics—the proposed approach provides the necessary discriminative signal to resolve these ambiguities. Consequently, this enables the model to look beyond the semantic content of the transcript and accurately identify nuanced emotional expressions that generic descriptors fail to capture.

Chapter 6

Conclusion

6.1 Conclusion

This research presented a novel, automated Multimodal SER framework designed to bridge the gap between controlled, acted speech and naturalistic, ”in-the-wild” communication. By integrating the generative reasoning capabilities of LLMs with a structured, quantitative acoustic representation, the proposed system systematically addresses the critical limitations of existing baselines—specifically their reliance on coarse-grained taxonomies, feature scarcity, and inefficient prompting mechanisms. To resolve the ”semantic bottleneck” where acoustically similar emotions become indistinguishable, we introduced a hybrid feature selection strategy incorporating up to 30 most discriminative descriptors from eGeMAPS. Simultaneously, we validated that transitioning from unstructured narratives to a Structured Key-Value Prompting strategy significantly mitigates information dilution, enhancing the model's ability to attend to these high-dimensional paralinguistic cues.

Experimental validation confirmed the efficacy of this approach across disparate domains. On the acted IEMOCAP dataset, the model achieved an **UA of 70.9%**, representing a **0.3% improvement** over the SpeechCueLLM baseline. On the more challenging, naturalistic MSP-Podcast corpus, the framework reached a **Macro-F1 score of 22.5%**, achieving a substantial **4.0% improvement** and demonstrating exceptional capability

in recognizing fine-grained secondary emotions such as *Concerned* and *Disappointed*.

We also evaluated the model's robustness to transcription errors, finding that the framework is highly resilient to textual noise; despite the high WER inherent to automatic transcription, the impact of ASR errors resulted in a degradation of only **3% absolute** on IEMOCAP, while conversely yielding a slight performance improvement on the naturalistic MSP-Podcast dataset. This confirms the system's viability for real-world deployment where perfect ground-truth transcripts are rarely available.

Ultimately, this work confirms that augmenting LLMs with precise, structured paralinguistic features provides a scalable and domain-agnostic solution for robust emotion recognition, paving the way for more empathetic and context-aware HCI systems.

6.2 Future Work

While the proposed framework establishes a strong baseline for LLM-based SER, several avenues for future research remain to further enhance performance and scalability.

6.2.1 Architectural Scaling with Next-Generation LLMs

The current implementation relies on **Llama-3-8B-Instruct**, which offers a balance of efficiency and performance. However, recent advancements in foundation models suggest that newer architectures (e.g., Llama-3-70B, GPT-4o) possess better reasoning capabilities. Future work should investigate whether replacing with newer models unlocks emergent abilities, such as the capacity to disambiguate complex mixed-emotional states without extensive fine-tuning.

6.2.2 Semantic-Aware Feature Selection (LLM-Select)

The current feature selection strategy relies on statistical methods (ANOVA/Lasso) that prioritize mathematical correlation over semantic interpretability. A promising direction is the adoption of LLM-Select [59], a methodology that uses the LLM itself to identify features that are most valuable for its specific reasoning process. This "semantic

selection" could reduce prompt bloat while ensuring that the retained acoustic tokens are maximally informative for the language model.

6.2.3 Probabilistic Multi-Label Prediction

Human emotion is rarely monolithic; natural speech often conveys a blend of overlapping affective states (e.g., Surprised yet Happy). The current framework frames SER as a single-label classification task, forcing the model to select one exclusive category. However, the MSP-Podcast dataset provides soft probability distributions across multiple emotions. Future iterations of the model could be trained to output a ranked dictionary of the top-3 probable emotions rather than a single tag. This multi-label approach would better reflect the ambiguity of human affect and provide downstream applications with a more nuanced understanding of the user's emotional state.

6.2.4 Weakly Supervised Learning with MSP-Conversation

Data scarcity remains a significant bottleneck for training robust SER models, particularly for naturalistic speech. The **MSP-Conversation** corpus [60] offers a vast repository of "in-the-wild" recordings that complements the MSP-Podcast dataset. Unlike the isolated, sentence-level segments of MSP-Podcast which often miss the temporal context of dialogue, MSP-Conversation consists of long-form interactions (10–20 minutes) annotated with time-continuous emotional traces for Valence (Negative to Positive), Arousal (Calm to Active), and Dominance (Weak to Strong).

A critical feature of this corpus is that its recordings are sourced from the same podcast pool as the MSP-Podcast database. This acoustic overlap is particularly advantageous for our proposed future work, as it ensures that representations learned from MSP-Conversation are directly transferable to MSP-Podcast without introducing domain shift artifacts. Furthermore, the corpus construction methodology explicitly balances emotional and demographic representation by selectively annotating segments known to contain diverse affective content.

To leverage this unutilized resource, future work could implement a Weak Supervision strategy. Since the corpus is primarily annotated with dimensional scores rather

than categorical labels, we propose developing a mapping function (either rule-based or clustering-based) to convert continuous VAD coordinates into pseudo-categorical emotion tags. This approach would allow us to amalgamate MSP-Conversation, IEMOCAP, and MSP-Podcast into a massive, unified training dataset. Such a large-scale pre-training regime has the potential to significantly improve the model's generalization capabilities across diverse acoustic environments while incorporating the long-term temporal dependencies essential for realistic emotion recognition.

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